

MA7202 Introduction to Functional Data Analysis 2025-26

Coursework 2

Student Number: 259039226

Task 1: Implement Functional Smoothing Techniques

i) The Darwin Sea Level Pressure (SLP) dataset contains monthly sea level pressure measurements recorded at Darwin, Australia, from 1951 to 2025 (75 years). Each year provides 12 monthly observations, so the data matrix is 12×75 (12 rows = months, 75 columns = years). As a preprocessing step, 1000 mb has been subtracted from every measurement, so values typically fall in the range 0–25 mb rather than 1000–1025 mb.

This dataset is treated as periodic functional data with each year's 12 monthly pressure readings form one complete annual cycle. This periodic structure makes the Fourier basis the most natural choice for smoothing. The Generalized Cross-Validation (GCV) method was used to find the optimal balance between fitting the data points and maintaining a smooth curve.

The GCV curve shows a L-shape with a lambda value of $\lambda = 10^0 = 1$. It indicates that the annual SLP cycle has enough inherent structure that a moderate amount of smoothing is required to remove monthly noise without losing the signal of climate events like El Niño.

```
# Loading the data
SLP_data = as.matrix(read.table("fda-SLP.dat"))
dim(SLP_data)
```

```
[1] 75 12
```

```
SLP_data = t(SLP_data)

# Identifying dimensions
nobs = 12
nyears = 75

#Task 1
#i)
# Creating fourier basis
timeRng = c(1, 12)
time = seq(1, 12, 1)
nbasis = 11
SLPbasis = create.fourier.basis(timeRng, nbasis)

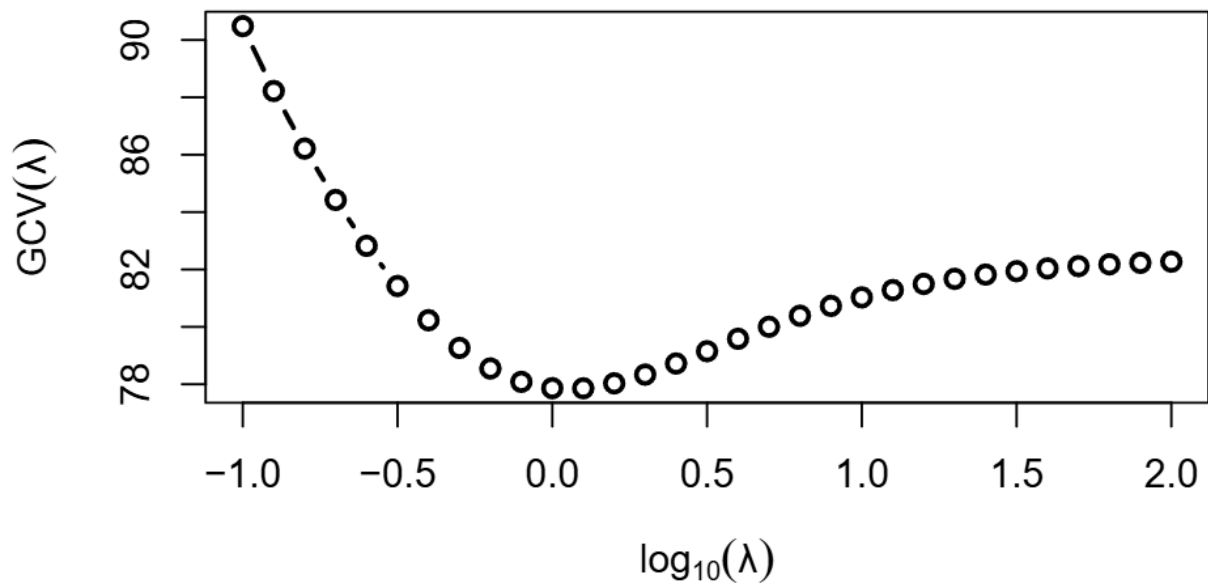
# Setting up the harmonic acceleration operator
harmaccelLfd = vec2Lfd(c(0, (2*pi/diff(timeRng))^2, 0), timeRng)

# Selecting Lambda using GCV
loglam = seq(-1, 2, 0.1)
nlam = length(loglam)
gcvsave = rep(NA, nlam)

for (ilam in 1:nlam) {
  lambda = 10^loglam[ilam]
  fdParobj = fdPar(SLPbasis, harmaccelLfd, lambda)
  smoothlist = smooth.basis(time, SLP_data, fdParobj)
  gcvsave[ilam] = sum(smoothlist$gcv)
}

# Plotting GCV to find the minimum
plot(loglam, gcvsave, type='b', lwd=2,
     main="GCV Criterion for Darwin SLP",
     xlab=expression(log[10](lambda)), ylab=expression(GCV(lambda)))
```

GCV Criterion for Darwin SLP



ii)

Using the selected lambda, the discrete monthly pressure values are transformed into continuous functional curves. The plot shows 75 separate curves with one for each year from 1951 to 2025. The curves span a pressure range of about 6 to 14 mb. The trough is around 6 to 7 mb in January and February and the peak is around 13 to 14 mb in July and August giving a seasonal amplitude of roughly 7 mb

In Darwin there is a consistent seasonal pulse where pressure is generally lower during the wet season between November and April and higher during the dry season between May and October. Years where the curves dip significantly lower than the average in the wet season correlate with the strong El Niño conditions.

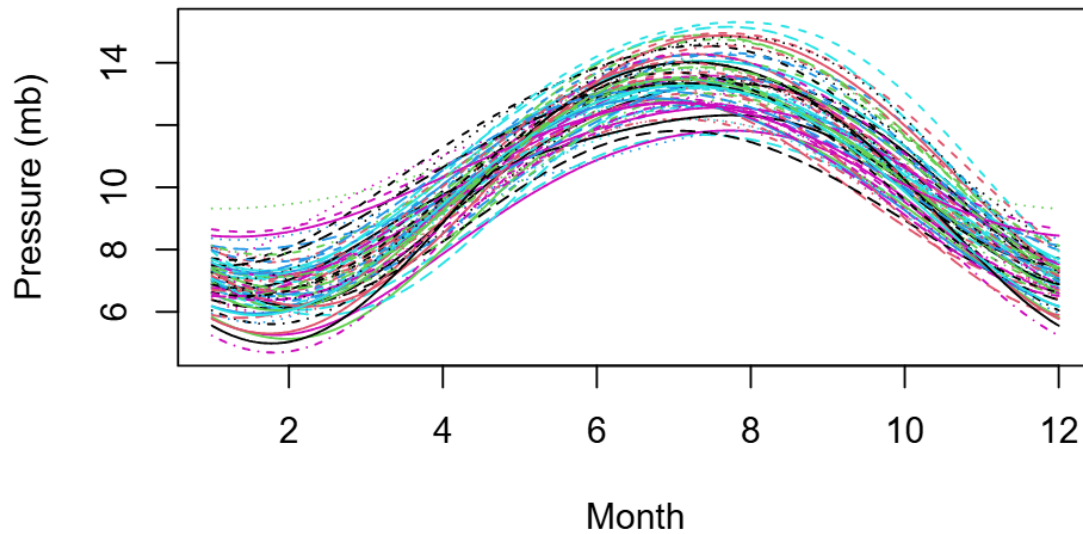
```
#ii)
# Lambda value based on GCV plot output
lambda = 1

SLP_fdPar = fdPar(SLPbasis, harmaccelLfd, lambda)
SLP_fit = smooth.basis(time, SLP_data, SLP_fdPar)
SLP_fd = SLP_fit$fd

# Adding descriptive names for the functional object
SLP_fd$fdnames = list("Month", "Year", "SLP relative to 1000mb")

# Plotting the smoothed functional data
plot(SLP_fd, lwd=1, main="Smoothed Darwin SLP Functional Data (1951-2025)",
      xlab="Month", ylab="Pressure (mb)")
```

Smoothed Darwin SLP Functional Data (1951–2025)

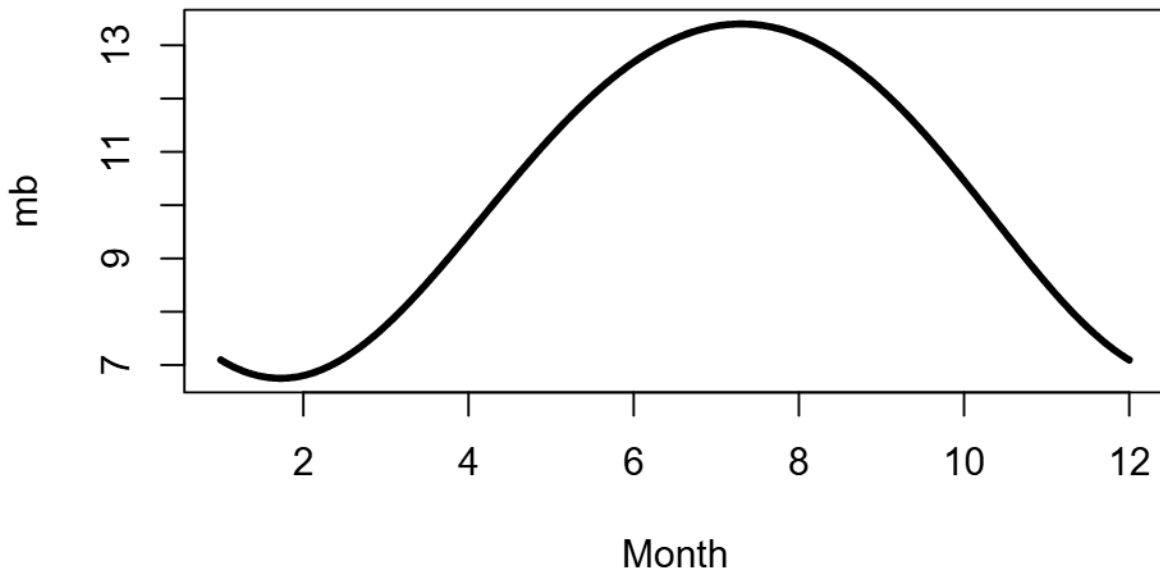


Task 2: Implement Descriptive Statistical Analysis

i) The mean function represents the average annual pressure cycle in Darwin over the 75-year period. It shows a peak of about 13.1 mb around July and August which corresponds to the winter and dry season and a trough of about 7.1 mb during January and February which corresponds to the summer and wet season. This gives a mean seasonal amplitude of approximately 6 mb, reflecting the dominant monsoonal pressure cycle at Darwin.

```
#Task 2
#i)
meanSLP = mean.fd(SLP_fd)
plot(meanSLP, lwd=3, main="Mean SLP Cycle", xlab="Month", ylab="mb")
```

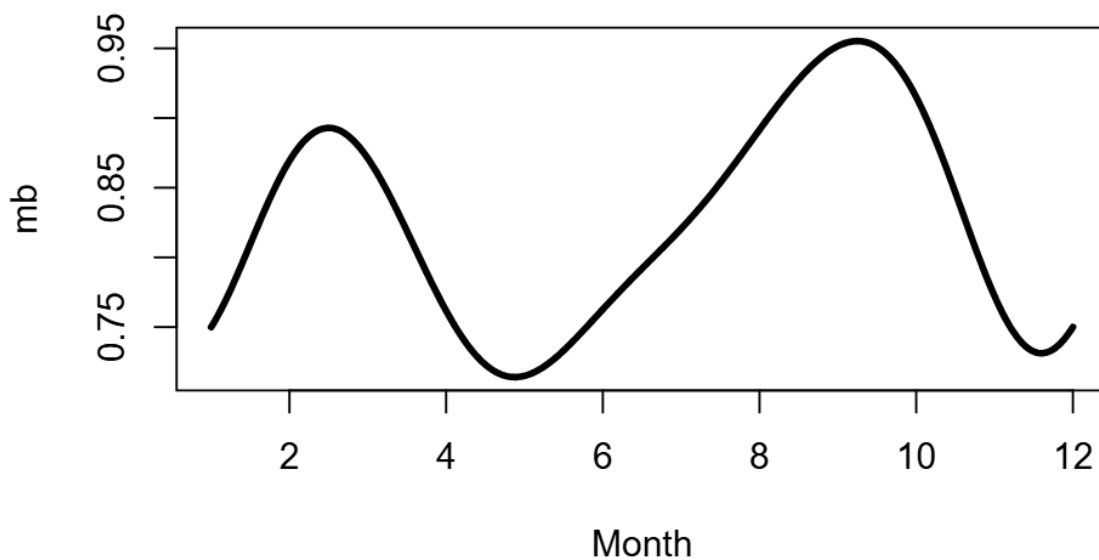
Mean SLP Cycle



```
stddevSLP = std.fd(SLP_fd)
plot(stddevSLP, lwd=3, main="Standard Deviation", xlab="Month", ylab="mb")
```

The standard deviation function indicates the changes in the pressure at different times of the year. The function has two distinct peaks. The larger peak of approximately 0.95 mb around October and a smaller peak of approximately 0.87 mb around February with a minimum of approximately 0.70 mb around June. The range of the standard deviation function is relatively narrow at about 0.25 mb which suggests that the shape of the seasonal cycle is consistent across years even if the absolute pressure level varies.

Standard Deviation

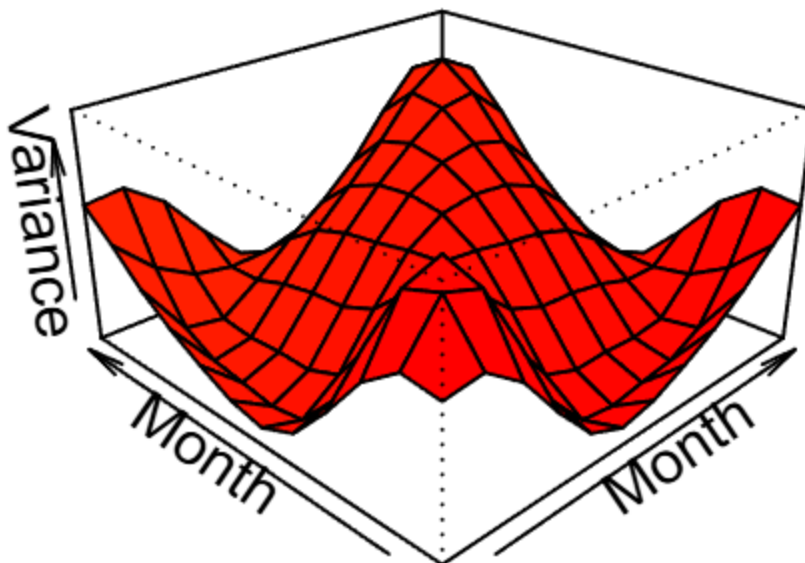


ii) The 3D surface plot illustrates how pressure at one time of the year relates to pressure at another. The z-axis represents the variance at each specific month. There are two distinct elevated ridges visible along the diagonal corresponding to the high-variance months at approximately months 2 and 10, consistent with the standard deviation function. A high covariance between early and late months suggests that if an El Niño event starts early in the year the pressure anomaly is likely to persist across consecutive months even though the non-diagonal values between opposing seasons show this persistence does not extend across the full annual cycle.

```
#ii)
SLPvar_bifd = var.fd(SLP_fd)
# Evaluate on a grid for plotting
SLPvar_mat = eval.bifd(time, time, SLPvar_bifd)

# 3D Perspective Plot
persp(time, time, SLPvar_mat,
      theta=-45, phi=25, r=3, expand=0.5,
      xlab="Month", ylab="Month", zlab="Variance",
      main="3D Variance-Covariance Surface", col=rainbow(5000))
```

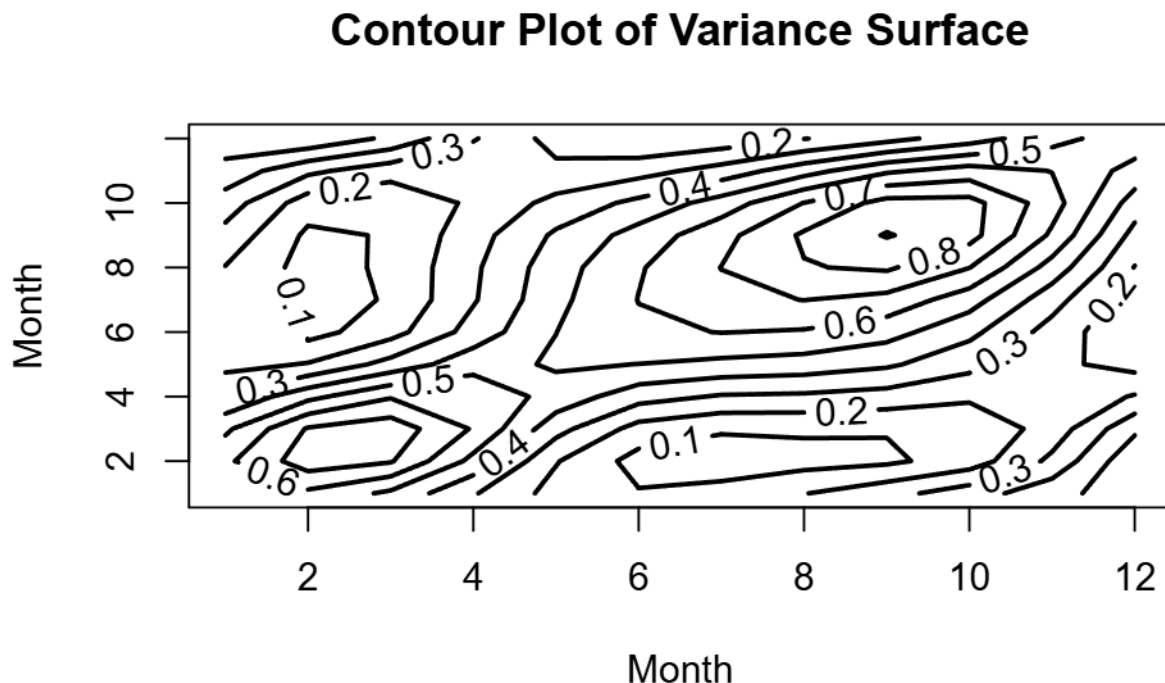
3D Variance-Covariance Surface



iii) The contour plot provides a top-down view to precisely locate the features seen on the 3D surface. The concentric rings identify the regions of maximum variance. The innermost contour in the bottom-left corner around months 1 and 2 reaches a value of 0.6 mb^2 while the innermost contour in the upper-right region around months 9 and 10 reaches 0.8 mb^2 , confirming that October and November pressure

is the most variable month across the entire 75-year record.

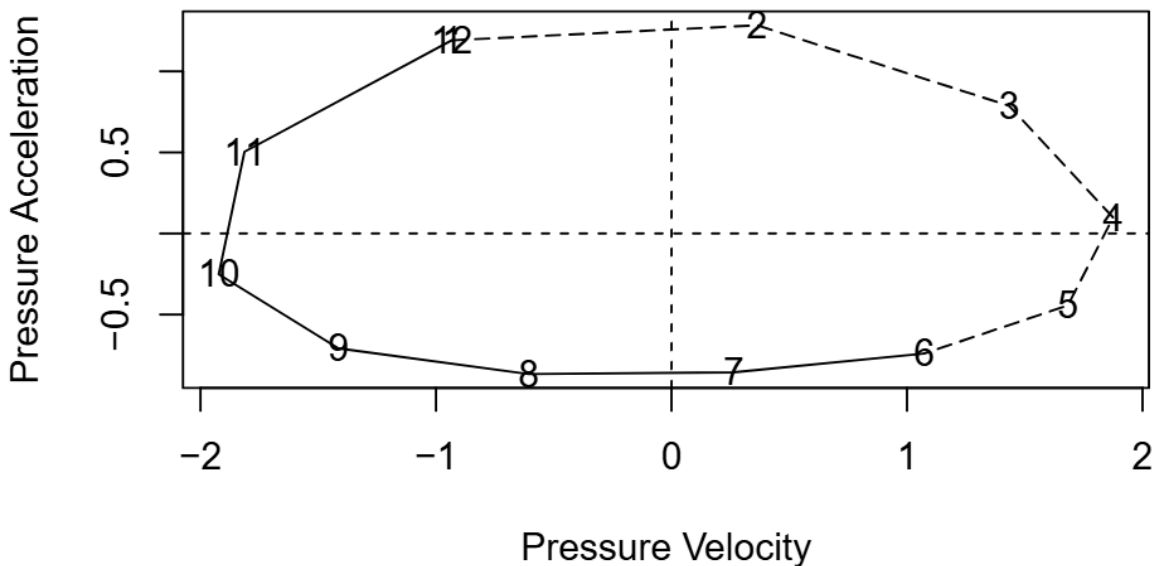
```
#iii)
# Contour Plot
contour(time, time, SLPvar_mat, lwd=2, labcex=1,
        main="Contour Plot of Variance Surface", xlab="Month", ylab="Month")
```



iv) The phase-plane plot graphs the velocity of the pressure against its acceleration. The circular motion represents the energy cycles which is the energy exchange within the annual cycle across the 12 month annual cycle. The loop is angular rather than smooth and elliptical with sharp corners indicating that Darwin's seasonal pressure transitions occur relatively abruptly rather than as a gradual sinusoid and reflects the sharp onset and withdrawal of the monsoon system. The loop shows the transition from the humid wet season to the cool dry season and closes near the starting point which indicates a stable repeating annual cycle.

```
#iv)
# Using the simplified method
phaseplanePlot(time, meanSLP, labels=list(evalarg=time, labels=1:12),
               xlab="Pressure Velocity", ylab="Pressure Acceleration",
               main="Phase-Plane Plot of Mean SLP")
```


Phase-Plane Plot of Mean SLP



Task 3: Implement Functional Principal Component Analysis

i) The scree-plot displays the eigenvalues in descending order. Eigenvalue 1 is about 4.5, eigenvalue 2 is about 2.0, eigenvalue 3 is about 1.1 and eigenvalues 4 to 6 are all approximately 0 to 0.1 lying very close to the fitted linear trend line. Eigenvalues 4 to 6 merge with the trend line indicating noise and on this basis three principal components are selected.

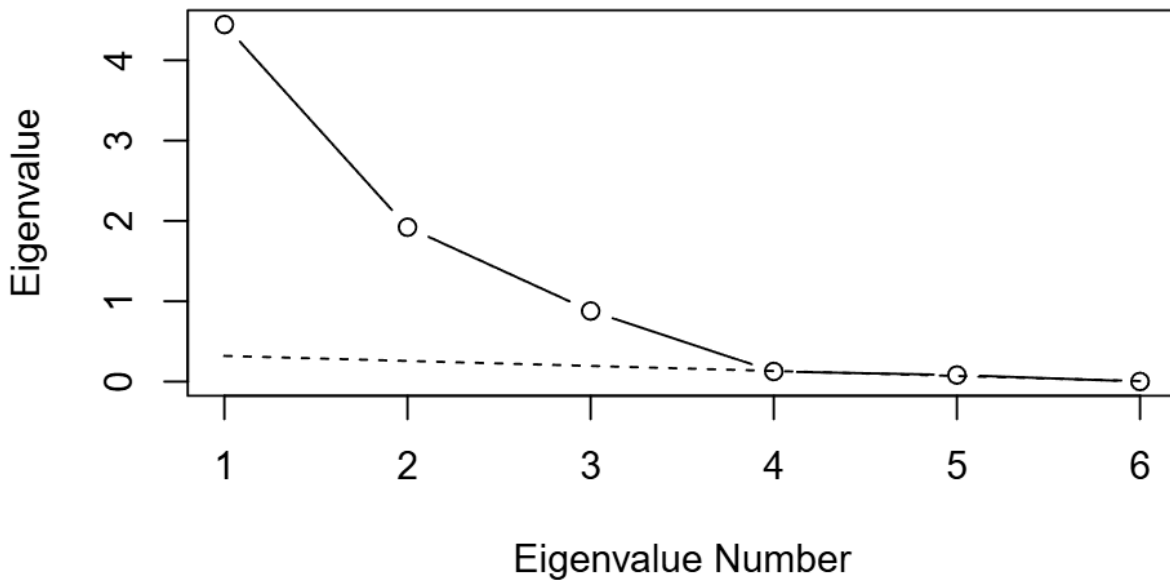
```
#Task 3
#i)
# Presetting the number of PCs or harmonics
nharm=4
SLP_pca_check = pca.fd(SLP_fd, nharm)

# Using the scree-plot to determine the number of PCs or harmonics.
pca_eig = SLP_pca_check$values
neig = 6 # Total eigenvalues to show

# Fitting a linear trend to the tail
x = matrix(c(rep(1, 3), 4:6), 3, 2)
y = as.matrix(pca_eig[4:6])
trend = lsfit(x, y, int=FALSE)$coef

# Plotting Scree-plot.
plot(1:neig, pca_eig[1:neig], type="b", main="Scree Plot for SLP",
     xlab="Eigenvalue Number", ylab="Eigenvalue")
lines(1:neig, trend[1] + trend[2]*(1:neig), lty=2)
```

Scree Plot for SLP



ii) Selecting three harmonics accounts for approximately 97.2% of the total variation specifically 59.6%, 25.8%, and 11.8% respectively which provides a high level of detail for the climatic model.

The first harmonic explains 59.6% of the total variability. The + signs in the plot are above the mean curve across the entire 12-month period which represents years where the sea level pressure was higher than average throughout the whole year. This variation captures the atmospheric shifts associated to years with high scores on this harmonic correlating with El Niño periods, where pressure in Darwin remains elevated regardless of the season.

The second harmonic shows 25.8% of the seasonal contrast whereby the + signs are mostly prominent during the middle of the year of months 5 to 9 which corresponds to the Australian dry season. This component captures years where the seasonal peak in pressure sharper than the average. A high score here indicates a year with a very pronounced contrast between the low-pressure wet season and a high pressure dry season.

The third harmonic captures 11.8% of meaningful variation. It appears as an oscillation where the + signs are higher at the start and end of the year which is the wet season but lower in the middle. It represents a phase shift variation accounting for years where the monsoon cycle arrived earlier or lasted longer than usual. It adjusts the shape of the transition between the seasons that the first two harmonics might miss.

```
#ii)

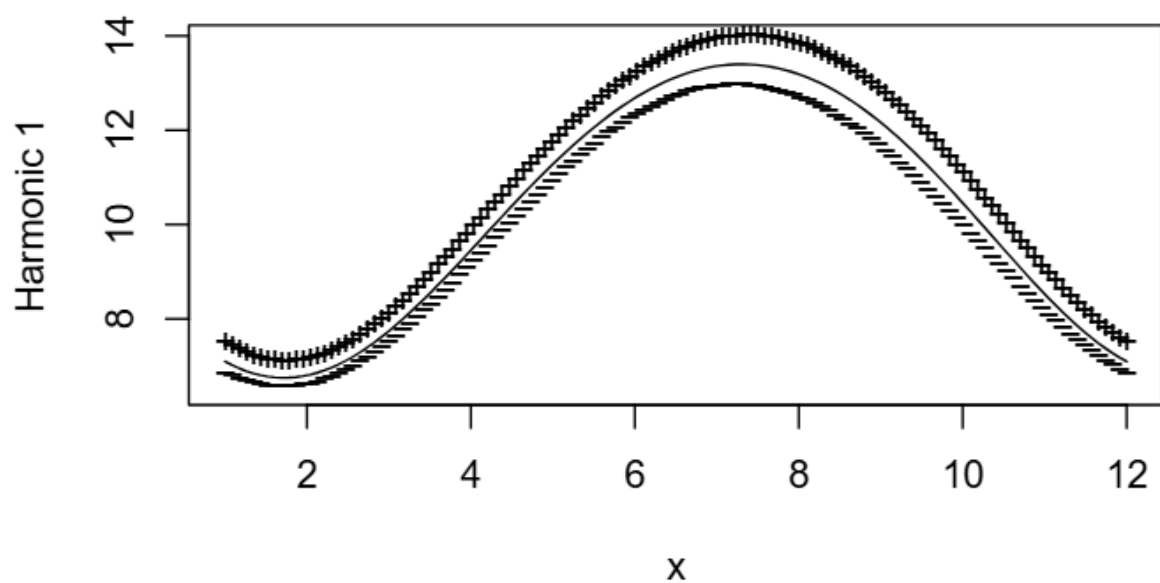
nharm = 3
SLP_pcalist = pca.fd(SLP_fd, nharm)

# Printing proportions
print(SLP_pcalist$varprop)
```

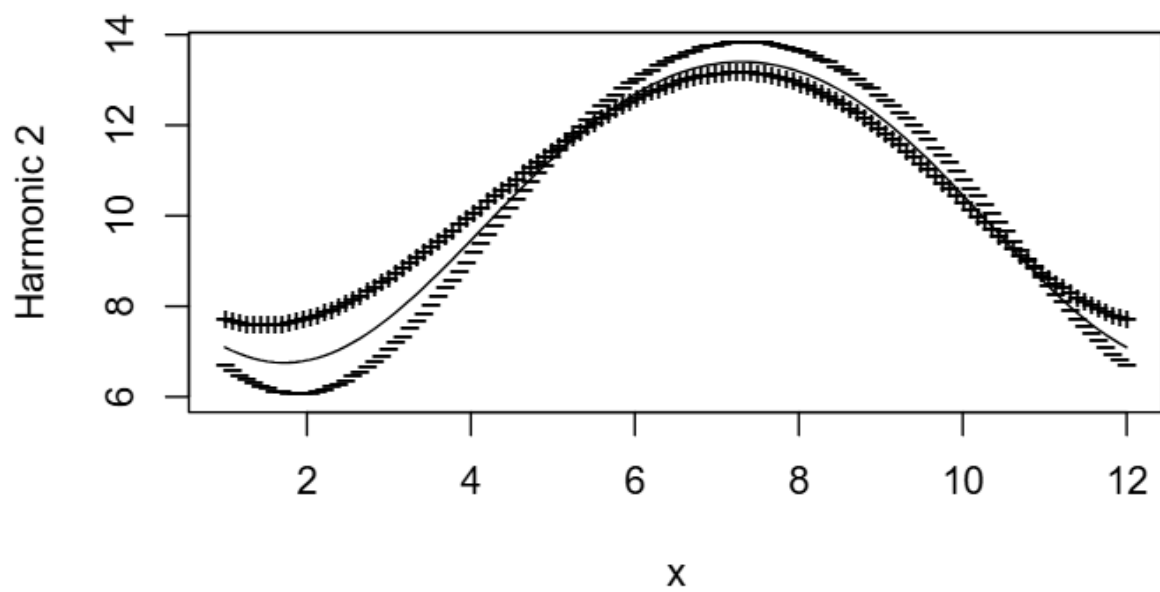
```
[1] 0.5960823 0.2577407 0.1178713
```

```
# Plotting harmonics
plot.pca.fd(SLP_pcalist, expand=1.5)
```

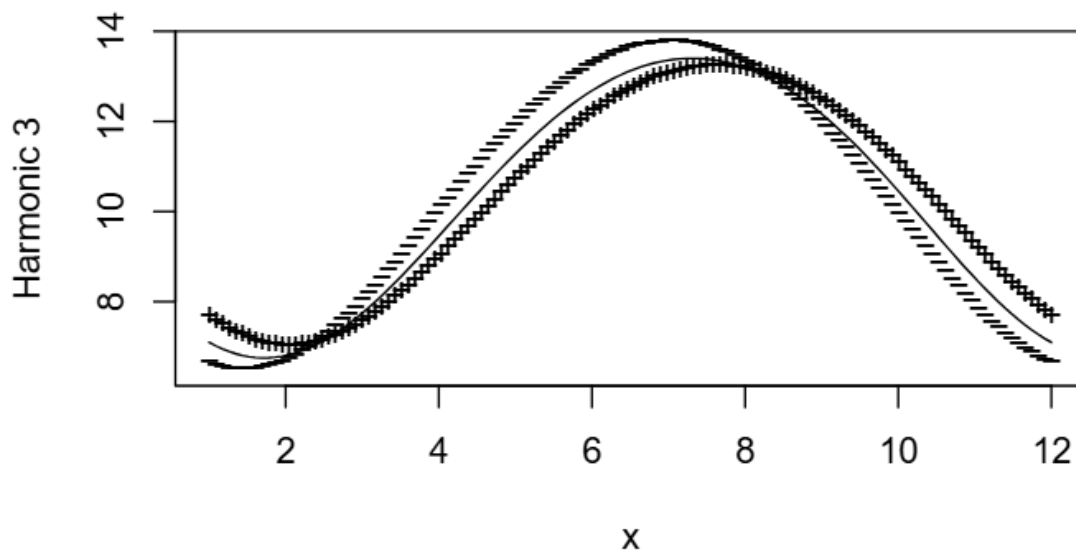
PCA function 1 (Percentage of variability 59.6)



PCA function 2 (Percentage of variability 25.8)



PCA function 3 (Percentage of variability 11.8)



iii) Rotation reallocates the variance so that each harmonic is more localised to a specific time of year, making them easier to relate to physical climate events. After rotation the variance is distributed more equally at approximately 35.4%, 29.6%, and 32.2% and these sum to 97.2% which exactly matches the total variation explained by the three unrotated harmonics.

The first rotated harmonic at 35.4% shows the largest separation between its + and – sign lines concentrated around months 10 to 12 with the + line sitting above the mean in those months while converging toward the mean during months 1 to 6. This localises the component to the late dry season and monsoon period of October to December and isolates the variation in the stability and timing of the winter high-pressure system.

The second rotated harmonic at 29.6% shows the – line significantly low at the wet-season trough, with the most notable separation between + and – occurring at months 1 to 3. This component captures variability in the wet season pressure minimum and identifies years with extreme low-pressure anomalies associated with heavy monsoon events or fully developed La Niña conditions in the Darwin region.

The third rotated harmonic at 32.2% shows the largest + sign concentrated at the mid-year peak around months 6 to 8 with the – line sitting well below the mean during this dry-season period while converging toward the mean at the start and end of the year. This isolates variability in the amplitude and strength of the winter dry-season high-pressure system allowing analysis of years where the dry-season pressure peak was unusually strong or weak in isolation from wet-season and transition-season fluctuations.

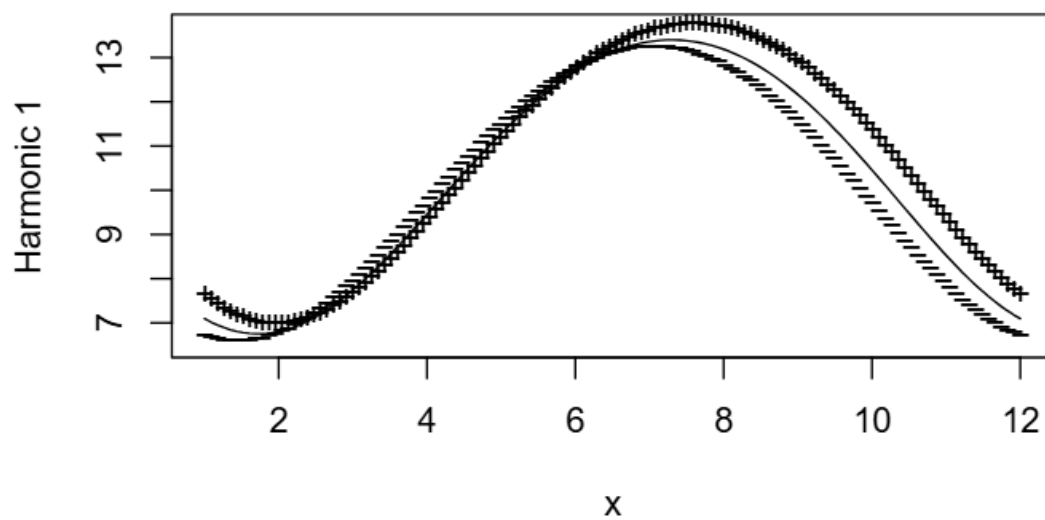
```
#iii)
SLP_rotpcalist = varmx.pca.fd(SLP_pcalist)

# Printing rotated proportions
print(SLP_rotpcalist$varprop)
```

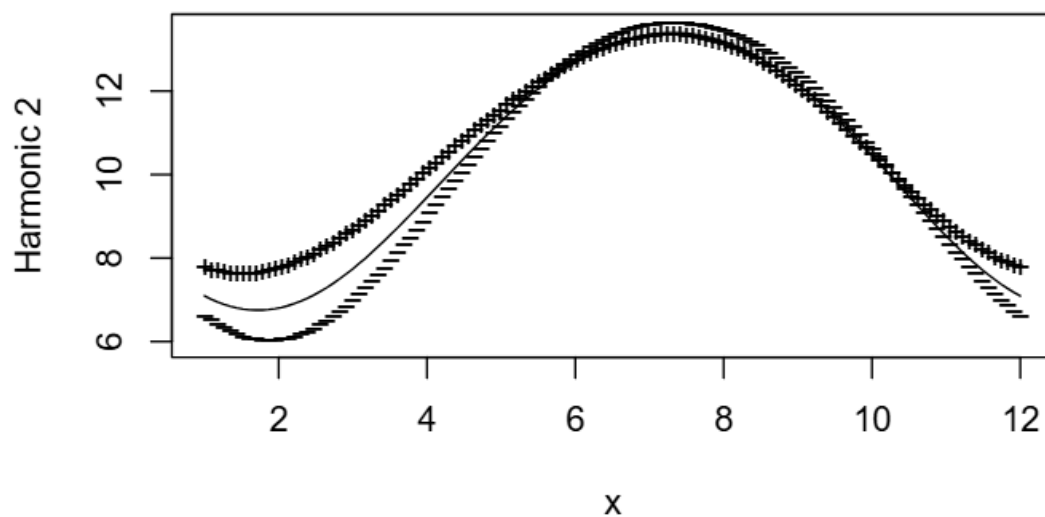
```
[1] 0.3541542 0.2958613 0.3216788
```

```
# Plot rotated harmonics
plot.pca.fd(SLP_rotpcalist, expand=1.5)
```

PCA function 1 (Percentage of variability 35.4)



PCA function 2 (Percentage of variability 29.6)



PCA function 3 (Percentage of variability 32.2)

